

Tech Stack				
IAM, VPC, S3, SageMaker, EC2, Lambda, EBS, CloudWatch (for general monitoring)				
Pipeline Setup				
AWS Service(s)	Design Choice	Alternative Option(s)	Justification	Estimated Cost
Identity Access Management (IAM)	Create IAM role, granting myself least-access privileges.	Build project from root AWS account.	IAM prevents potential misuse and adheres to cyber-security best practices.	\$0.00
Virtual Private Cloud (VPC)	Create private subnet and endpoints for S3 Buckets.	All services run in AWS' default VPC and communicate over the internet.	Using a private subnet is more secure, best practice to avoid exposure to internet given no service needs access to it.	\$0.00
Data Curation and Preprocessing				
AWS Service(s)	Design Choice	Alternative Option(s)	Justification	Estimated Cost
Simple Storage Service (S3)	Download data from ArXiv and upload to S3 "Data" Bucket.	N/A	Very simple and cheap to store data in S3. EC2, SageMaker, and Lambda will all need to access the same data at different stages.	\$1.57 per month
SageMaker	Analyze raw data in SageMaker notebook (also used later and for pre-training).	Use small or medium EC2 instance instead to analyze data.	SageMaker offers competitive prices and Jupyter Notebooks optimized for DL.	\$64.92 per month (shared cost)
Elastic Compute Cloud (EC2) and Elastic Block Storage (EBS)	Pre-process the data with basic PyTorch transformations on a small EC2 instance with a gp3 EBS attached volume.	Use SageMaker Notebook for pre-processing, pre-process on local machine, or with Lambda.	SageMaker could be a slightly cheaper option than EC2, but a goal of this project is to gain significant AWS experience and SageMaker is being used for other parts of the project. Same justification as above for local machine. Lambda is	\$17.58 per month (shared cost)

			serverless but would have to invoke too many Lambda function calls compared to EC2.	
S3	Store pre-processed data in “data” S3 bucket, using prefix to differentiate from raw data.	Store pre-processed data in a new S3 Bucket.	Adding another bucket creates more infrastructure overhead. S3 is highly scalable, so storage limits are not an issue.	\$1.57 per month
SageMaker	Check pre-processed data for any errors before pre-training, in same SageMaker Notebook.	Skip this step.	Validates quality of the training data before spending time and incurring pre-training costs.	\$64.92 (shared cost)
Training				
AWS Service(s)	Design Choice	Alternative Option(s)	Justification	Estimated Cost
SageMaker and S3	Train diffusion model using SageMaker Notebook with “ml.g5.xlarge” spot instance (CUDA-accelerated GPU). Save weights periodically to “models” S3 Bucket.	Use EC2 instance instead of SageMaker.	SageMaker is a cheaper alternative to EC2 as it’s optimized for ML training. Because spot instance will likely terminate during training, saving the weights after each epoch ensures that the disruption to training is minimized.	\$64.92 (shared cost) for SageMaker. \$0.47 for S3 (shared cost)
S3	Save final model to “models” S3 Bucket.	N/A	Saves the final model for future use.	\$0.47 (shared cost)
Inference				
AWS Service(s)	Design Choice	Alternative Option(s)	Justification	Estimated Cost
Lambda	Pre-process inputs using a Lambda function.	Use small EC2 instance, like with the pre-processing step, or use local machine.	Lambda is serverless, cheap, and lightweight for simple pre-processing, which makes it ideal for a	\$0.00 for 1,000 calls per month

			few function calls, compared to EC2. I want experience configuring Lambda over using local machine.	
EC2	Run inference on pre-processed testing samples, using the same EC2 instance as for pre-processing.	Use SageMaker Notebook instead and posting the image to a website.	SageMaker charges for endpoints, even when the model is in idle, unlike EC2. Instead	\$17.58 per month (shared cost)
S3 and EC2	Save model image outputs to “outputs” S3 Bucket and generate & return temporary pre-signed URL(s) to visualize model output(s) stored in “outputs” bucket	Sending the image to the user without saving the result and/or posting the image to a website.	Model outputs should be saved somewhere. Much more expensive and less secure to post to a website as opposed to providing a secure link to the image inside of the “outputs” S3 Bucket.	\$17.58 per month (shared cost) for EC2. \$0.92 for “outputs” S3 Bucket.
Post-Training Cleanup				
AWS Service(s)	Design Choice	Alternative Option(s)	Justification	Estimated Cost
S3	Delete “data” S3 Bucket. Delete all models except for last model in “models” Bucket.	Don’t delete.	Not deleting unused data incurs unnecessary costs.	\$0.00
SageMaker	Delete Notebook used for analysis/visualization and training.			\$0.00
Post-Project Cleanup				
AWS Service(s)	Design Choice	Alternative Option(s)	Justification	Estimated Cost
S3	Download final model and generated outputs to local computer. Then delete “models” and “outputs” Buckets	Don’t download and delete.	Not deleting unused data incurs unnecessary costs but the model and its data should be	\$0.00

Lambda	Delete Lambda Function used for inference pre-processing.		preserved for future use.	\$0.00
EC2	Delete pre-processing and inference EC2 Instance.			\$0.00
VPC	Delete the VPC and all associated resources, including subnets, route tables, and VPC endpoints.			\$0.00
All	Confirm deletion of all cost-incurring project infrastructure. If any remains, delete it.			\$0.00